Name: _____ Period: _____

Directions: Each sample answer below would lose credit. First, figure out exactly what the question asks for. Then find the specific place where the student went wrong, and fix it. Use your ESSRT.

Unit 2 Practice: Where Did It Go Wrong?

Question 1

Hikers in the Adirondack Mountains often see large rocks split by wide cracks. In late fall, rainwater fills the cracks during the day. At night, the temperature drops below 0°C and the water freezes. This happens over and over for many years.

Explain how freezing water makes the cracks in the rock wider. Use a property of water in your answer.

Sample answer:
The cracks get wider because the water freezes over and over for many years.

What does the question ask for? List every part.	**Where did the student go wrong?** Be specific.	**Fix it:** Write the part that is missing or wrong.

Question 2

A student found a light-colored rock made of large crystals of three different minerals. The student tested the pink mineral in the rock and recorded these observations: pink with a glassy luster; scratches a glass plate; breaks along two flat surfaces that meet at 90°; no lines on the flat surfaces.

Use the Mineral Identification Flowchart to identify the pink mineral. Then state one observation that helped you identify it.

Sample answer:
Mineral: granite
Observation: It breaks along two flat surfaces that meet at 90°.

What does the question ask for? List every part.	**Where did the student go wrong?** Be specific.	**Fix it:** Write the part that is missing or wrong.

Question 3

In 2018, lava from Kīlauea volcano in Hawai'i poured out onto Earth's surface and cooled quickly. The rock that formed is dark colored and high in iron and magnesium. Its crystals are too small to see without a magnifying glass.

Use the ESSRT to identify the rock. Then explain why its crystals are so small.

Sample answer:
Rock: basalt
Explanation: It is dark colored and high in iron and magnesium.

What does the question ask for? List every part.	**Where did the student go wrong?** Be specific.	**Fix it:** Write the part that is missing or wrong.

Question 4

A student heated a beaker of water from the bottom center and added a drop of food coloring. The colored water rose up the middle of the beaker, moved across the top, and sank along the sides.

*Describe the movement of the water in the beaker. Then explain why the heated water rises. Use the word **density** in your explanation.*

Sample answer:
The water moves in a loop. It goes up in the middle, across the top, and down the sides. This is called convection.

What does the question ask for? List every part.	**Where did the student go wrong?** Be specific.	**Fix it:** Write the part that is missing or wrong.

Question 5

A student tested two white minerals and recorded these observations.

Sample	Observations
X	White; bubbles when a drop of acid is placed on it; scratched by a glass plate; breaks into pieces with flat sides
Y	White; does not react with acid; scratches a glass plate; breaks with curved, uneven surfaces

*Identify **two different** properties from the data table that could be used to tell Sample X and Sample Y apart.*

Sample answer:
1. Sample X bubbles when acid is placed on it.
2. Sample Y does not react with acid.

What does the question ask for? List every part.	**Where did the student go wrong?** Be specific.	**Fix it:** Write the part that is missing or wrong.